

Ryosei Takiguchi



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PROFILE

MSc Robotics student in the JEMARO program (Keio University & École Centrale de Nantes) with a specialization in autonomous driving and robotic manipulation. My research focuses on visual haptic integration for contact-rich tasks. Having conducted research in both Japan and France, I possess strong cross-cultural adaptability and a proven ability to collaborate in international projects. My ultimate research goal is to advance Human-Robot Interaction, developing intelligent systems where robots and humans can coexist and collaborate seamlessly in real-world settings.

EDUCATION

09/2024 – today	Master's double degree “Japanese European Master on Advanced Robotics – JEMARO” - Keio University (M2) - École Centrale de Nantes (M1) - GPA to Date: 3.35 (Scale: 0.0 = lowest, 4.0 = highest)
04/2020 – 03/2024	Bachelor's degree “B.Eng. in System Design Engineering” - Keio University - Graduated with GPA: 3.07 (Scale: 0.0 = lowest, 4.0 = highest)
04/2017 – 03/2020	High School Diploma (General Course) Kanazawa University Senior High School

RESEARCH & ACADEMIC PROJECTS

01/2026 – today	Visual Haptics Manipulation: Robotic Drawer Opening - Implement a visual-haptics manipulation pipeline enabling a robotic arm to perform contact-rich tasks through force-feedback and visual perception. - Robotic Manipulation, Haptics, Force Control, Human-Robot Interaction.
09/2025 – 01/2026	Object-Triggered 3D Semantic SLAM - Developed an efficient 3D Semantic SLAM system that selectively reconstructs environments based on object detection to optimize computational resources for mobile robots. - SLAM, Computer Vision, Perception.

04/2025 – 09/2025

Semantic 3D Mapping

- Built a semantic mapping framework combining 2D segmentation and 3D SLAM using RGB-D data. Enhanced environment understanding for autonomous robots through semantic 3D reconstruction.
- SLAM, Semantic Segmentation, Computer Vision.
- Paper accepted by AMC 2026

06/2025 – 07/2025

Autonomous Driving : Obstacle avoidance

- Implemented a LiDAR-based obstacle detection pipeline on a real autonomous vehicle, including filtering and clustering. Integrated detection results with the ROS2 navigation stack for autonomous path planning.
- Path planning, Navigation, Autonomous vehicle.

PROFESSIONAL EXPERIENCES

12/2025 – 12/2025

TIER IV Inc. / Autonomous Driving workshop

- Implemented planning modules of Autoware and developed re-start function from middle of the road.

08/2025 – 08/2025

Nissan Motor Co., Ltd. / AD/ADAS Development Internship

- Assisted in the development and evaluation of safety critical ADAS software layers.

QUALIFICATIONS & SKILLS

Programming Languages	Python, C++, MATLAB/Simulink, Java Script
Robotic Tools	ROS2, Isaac Sim, Isaac Lab, Git, PyTorch
Computer Vision	OpenCV, Open3D, Point Cloud Library
Method & Theory	SLAM, Reinforcement Learning, Control Theory, Machine Learning.
Languages	Japanese - Native English - C1 French - A2

PUBLICATIONS

- R. Takiguchi and T. Nozaki, "Dense SLAM-Based 3D Semantic Mapping via Pose Completion and Confidence-Weighted Multi-View Fusion," in *Proceedings of the IEEE International Conference on Advanced Motion Control (AMC)*, Daegu, South Korea, Mar. 2026. (Accepted)
- R. Takiguchi and T. Nozaki, "Estimated Speed Compensation Based on Two Types of Axis Errors," in *Proceedings of the 10th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Kyoto, Japan, Mar. 2024.